

Postactivation Potentiation and the Asynchronous Action of Muscular and Neural Responses

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Purpose: This study examined the underlying mechanisms of postactivation potentiation and the time course of muscular- and neural-related variables. **Methods:** Fourteen trained males executed 4 sets of six 6-second maximum isometric conditioning plantar flexions, with 15 seconds and 2 minutes of interval between the contractions and sets, respectively. Peak twitch torque (TT), rate of torque development, time to peak torque, half relaxation time, and the neural-related variables of H-reflex and electromyogram, normalized to the maximum M-wave (H/M and RMS/M, respectively), were evaluated, as well as the level of the voluntary activation, assessed by the twitch interpolation technique. All neural-related variables were analyzed for the trial within each set when TT was maximal and for the trial within each set when the neural-related variable itself was maximal. **Results:** Compared with the baseline measures, TT and rate of torque development significantly increased in all sets ($P < .001$), whereas time to peak torque and half relaxation time significantly decreased in sets 1 to 4 and 2 to 4, respectively ($P < .001$). However, H/M and the RMS/M did not change for the repetition of each set for which the TT was maximal ($P > .05$). Interestingly, the within-set maximum H/M ratio of the lateral gastrocnemius muscle revealed a significant increase in all sets ($P < .05$), compared with the baseline measures. **Conclusion:** One set of 4 contractions with 6-second duration is sufficient to cause postactivation potentiation for most participants, whereas peak TT augmentation does not coincide with changes in the examined neural-related variables. Further experiments should consider the time lag on their maximal values and their inherent between-participants variability.

Keywords: electromyography, H-reflex, M-wave, twitch torque, fatigue

Contraction history may have an acute positive or negative impact on the motor output of skeletal muscles. One well-established phenomenon that may increase force production is postactivation potentiation (PAP), defined as an increase in muscle performance after an evoked or voluntary conditioning stimulus.¹ The potentiation in force is expressed both in performance² and electrically induced muscle activation.³ In the latter case, twitch torque (TT) potentiation can be recorded in vivo as an increase in the electrically evoked peak isometric torque after a conditioning stimulus.⁴ The most acknowledged mechanism for the existence of TT potentiation at molecular level is the phosphorylation of myosin regulatory light chains, which is suggested to enhance the excitation–contraction coupling efficiency.¹ Although this mechanism is located at the muscular level, it cannot be ruled out that other additional peripheral (muscular) or even central (neural) mechanisms coexist and may facilitate or impede the occurrence of PAP under different circumstances.

The investigation of neural contributions to the TT potentiation after an intense voluntary muscle contraction is limited. A few studies tested the excitability of α -motor neurons through the Ia afferents after a voluntary contraction, with the traditional measurement of the Hoffman reflex (H-reflex).⁵ It has been suggested that a neural adaptation and particularly an increase in the recruitment of higher threshold motor units could partly explain the TT

potentiation. More specifically, it has been reported that the soleus (SOL),^{6,7} lateral gastrocnemius (LG),^{6,7} and vastus medialis⁸ H-reflex increases after a conditioning stimulus. However, there are some limitations related to the studies mentioned above. First, H-reflex amplitude might be influenced by neuromuscular transmission across the neuromuscular junction and by the action potential propagation across muscle fibers. Therefore, it is recommended to normalize it by the maximum M-wave (M_{max}).⁵ However, this practice was not always followed,⁶ and therefore changes in the H-reflex do not solely reflect neural changes.⁹ Second, the temporal occurrence of the TT and H-reflex potentiation in the studies mentioned above is different. More specifically, the maximum TT potentiation was observed within a short (<1 min) time after the conditioning contraction, whereas the H-reflex potentiation appears within a window of 3 to 11 minutes.^{6–8} Additionally, since TT and not the H-reflex amplitude is the variable considered to reflect the magnitude of PAP, one could argue that the potentiation of TT and H-reflex are not necessarily related in a manner of cause and effect. Therefore, if an increase in H-reflex (indicating higher α -motoneuron excitability via the Ia afferent pathway) could be one of the reasons for PAP, this should occur simultaneously with TT potentiation. In contrast to TT, studies have shown that H-reflex exhibits a strong inhibition instead of facilitation, immediately after the conditioning contraction.^{7,10} This could be attributed to post-activation depression,¹⁰ a spinal presynaptic mechanism that might mask the presence of any H-reflex potentiation, especially when H-reflex is measured with no background electromyographic (EMG) activity.¹⁰ This dictates the imperative need for further studies in the relationship between PAP and neural changes, as highlighted by a recent review.¹¹

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Considering all the above, the current study aimed to identify the peak response of muscular and neural mechanisms, in a PAP protocol consisted of 4 sets, with 6 subsequent conditioning maximal voluntary isometric contractions (MVICs) for each set. We hypothesized that the time course of muscle- and neural-related variables would demonstrate different behavior in magnitude of change, in the course of successive conditioning contractions. For this reason, the neural components were analyzed both for the trial within each set displaying maximal TT and for the trial within each set where any of the neural-related variable was maximal.

Methods

Subjects

Fourteen trained adult male handball athletes (age: 22.6 [2.4] y, height: 180.3 [5.9] cm, body mass: 72.3 [37.9] kg, training background: <10 y) were recruited from local sport clubs and participated voluntarily in the present study. The aforementioned characteristics (ie, male, trained participants) were chosen to favor PAP responses, as supported by previous literature.¹² Sample size (n = 14) was estimated by ensuring 85% power to detect an effect size of 0.35 at an alpha level of .05. They were free of any medical history related to lower-limb injury or any neuromuscular disorder. The study was approved by the local Ethics Committee on human research of the Aristotle University of Thessaloniki (ERC-006/2021) and was conducted according to the Declaration of Helsinki.

Design and Methodology

Experimental Procedures

This study used a repeated measurement design. One week before the day of evaluation, the participants visited the laboratory for their familiarization with the techniques and methods of the study. During the evaluation session, each individual started with a standardized 10-minute warm-up consisted of jogging, dynamic stretching exercises, and low-intensity jumps. After the warm-up, the EMG electrodes were placed over the SOL and LG muscle of the dominant limb. The participants were positioned on the dynamometer chair and performed 3 unilateral plantar flexion MVICs, with 6-second duration and 3 to 5 minutes of rest between the

repetitions. The participants were verbally encouraged and were instructed to apply as fast as possible their maximal force and hold on for up to 6 seconds. During the evaluation, visual feedback of the torque was provided to the participants through a monitor. During the next 10 minutes, the participant rested passively,¹³ while the examiner positioned and fixed the stimulation electrodes and adjusted the stimulation intensity for the H-reflex and $M_{\max}$ recordings. This time interval was also necessary to washout any PAP effect that could accumulate to the evaluation period that followed.¹⁴

After resting, baseline measurements were assessed for 5 minutes. During this period, the $M_{\max}$ was captured every 30 seconds.³ The H-reflex was measured 10 and 20 seconds after the $M_{\max}$ stimulus, during a tonic muscle activation at 5% of MVIC (Figure 1), which was sustained with the help of visual feedback from a monitor. Baseline measurements were followed by 4 sets of 6 conditioning plantar flexion MVICs,⁴ with 6-second duration, 2-minute rest between the sets and 15-second rest between each MVIC. When the torque had reached a plateau (~3 s after the onset of MVIC), a single supramaximal stimulus for the $M_{\max}$ was applied (superimposed twitch). No doublet was used to limit discomfort. A second supramaximal stimulus was delivered 2 seconds after the end of the MVIC, at rest (control twitch). A submaximal stimulus to evoke the H-reflex was applied 10 seconds later, while the participant executed a voluntary contraction at 5% of MVIC (Figure 1).

The supramaximal stimulation during the MVIC was used to estimate the level of voluntary activation (VA), using the interpolated twitch technique.¹⁵ The VA was calculated according to the formula suggested by Merton,¹⁶

$$VA_i = \left[1 - \left(\frac{TT_i^{\text{sup}}}{TT_i^{\text{con}}} \right) \right] \times 100,$$

where TT_i^{sup} and TT_i^{con} is peak of the superimposed and control twitch torque during and after the i th MVIC, respectively.

The supramaximal stimulation during rest was used to evaluate the TT, the rate of torque development (RTD), time to peak torque (TtP), and $M_{\max}$. The peak TT was measured as the maximum torque value,^{8,10} while the average RTD was calculated by dividing the peak TT with the time to reach the peak TT.^{17,18} The TtP was calculated as the interval between the onset of the TT and the peak

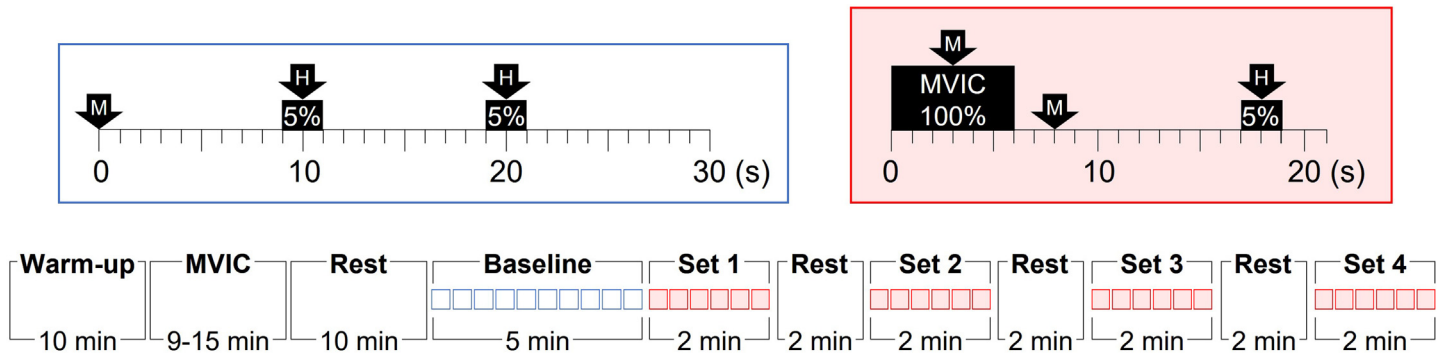


Figure 1 — Schematic description of the experimental protocol timeline, including warm-up, MVIC test, baseline measurements, and the main protocol. The enlargement in the box at top left describes a single trial (1 square) from the block of baseline measurements before the conditioning sets. Similarly, the shaded box at top right describes a single trial (1 square) from the block of conditioning sets. Downward arrows indicate with the respective letter the instance of stimulation with supramaximal intensity for the recording of $M_{\max}$ (M) or with submaximal intensity for the H-reflex recording (H). The boxes across the bottom half of the diagram designate the duration of maximal (100%) and submaximal (5%) MVIC. $M_{\max}$ indicates maximal M-wave; MVIC, maximal voluntary isometric contraction.

TT.¹⁴ The half relaxation time (HRT) was the interval between the time of peak TT until TT fell to the half of its maximum.^{3,19}

The repetition with the highest peak TT within each set was analyzed. For the neural-related variables, the repetition with the maximum value from each set was analyzed as well^{6,8} and peak TT was analyzed for these repetitions too. All baseline measurements, 5 minutes before the implementation of the conditioning stimuli (Pre) were averaged.

Torque Assessment

An isokinetic dynamometer was used (Cybex Norm, Cybex) to assess the isometric torque. Participants laid in prone position with their hip and knee joints fully extended (anatomically neutral position) and the angle between the shank and the foot was set at 90°.²⁰ The upper limbs rested on the sides of the dynamometer chair, while the hips and nondominant foot were fixed with nonelastic Velcro straps, firmly enough to minimize any movement of the foot relative to foot adapter and not to induce any numbness.

Electrical Stimulation

Electrical stimulation was delivered by a stimulator (Model S88, Grass Instruments, AstroMed, Inc) with bipolar surfaces electrodes. The tibial nerve was stimulated as described previously.¹⁰ The recordings of the H-reflex was set to produce its maximum value in SOL muscle, at an intensity of the M-wave threshold.²¹ The H-reflex for the LG was recorded at the same stimulation intensity. The intensity of the supramaximal stimulus for the $M_{\max}$ recording was 1.2 times greater than the intensity that the M-wave peak-to-peak amplitude did not increase any further.

Electromyography

The EMG signal of SOL and LG was recorded using a 2-channel BioNomadix wireless transmitter (Biopac Systems Inc). Bipolar disposable surface electrodes (Ag-AgCl), with 1 cm diameter, were applied according to SENIAM (Surface Electromyography for the Non-Invasive Assessment of Muscles) recommendations.²² The common reference electrode was placed over the tibia at the height between the stimulation and the recording sites. Prior to electrode placement, the skin was shaved and thoroughly cleaned with an alcohol solution, to keep skin resistance below 2 k Ω . The EMG signals were amplified $\times 1000$. All signals (stimulator trigger output, torque, and EMG) were digitized at 2-kHz sample rate and 16-bit resolution with the MP150 A/D converter (Biopac Systems Inc). All the data were stored for analysis with custom Labview (version 11.0f2, National Instruments) and MATLAB (MATLAB R2009b, MathWorks Inc) scripts. The EMG signal during MVIC was quantified calculating the root mean square (RMS) using a window of 20 ms.²³ The mean RMS was calculated over a 500-ms window, before the $M_{\max}$ stimulus. RMS values were normalized to the $M_{\max}$ peak-to-peak amplitude, measured from the same muscle at a relaxed state (RMS/ M_{SOL} and RMS/ M_{LG}). The peak-to-peak amplitude was analyzed for the H-reflex and $M_{\max}$, and the H-reflex amplitude was normalized to the $M_{\max}$ (H/ M_{SOL} and H/ M_{LG}). Since H/ M_{SOL} , H/ M_{LG} , RMS/ M_{SOL} , RMS/ M_{LG} , and VA are determined at least partly by inherent neural factors, they were characterized as neural-related variables.

Statistical Analysis

The mean (SD) was calculated for all dependent variables (peak TT, RTD, TtP, HRT, SOL $M_{\max}$, LG $M_{\max}$, H/ M_{SOL} , H/ M_{LG} ,

RMS/ M_{SOL} , RMS/ M_{LG} , and VA). Differences between sets (Pre, Set1, Set2, Set3, Set4) were examined for all the variables except the VA, which was analyzed during Set1, Set2, Set3, and Set4 only (no pre measurements were assessed). One-way analysis of variance with repeated measurements was used to examine the effect of time. In the case of significant main effect, Tukey post hoc test was performed to determine the existence of potentiation (comparison of pre vs Set1–4) or fatigue (comparison Set1 vs Set2–4). Hedges g effect size (Cohen d corrected for sample size less than 20), was calculated for pair-wise comparisons,²⁴ and values were interpreted as negligible (lower than 0.2), small (0.2 or lower than 0.5), moderate (0.5 or lower than 0.8), or large (larger or equal than 0.8). Statistical significance α was set at .05. All statistical analyses were performed using Statistica (version 8.0, StatSoft Inc) and R software (version 4.2.1, R Foundation for Statistical Computing).

To compensate for the asymmetric scaling of negative (decrease) and positive (increase) changes relative to the pre values, logarithmic conversion was applied according to the following formula²⁵:

$$\Delta x\% = \log_e(x_1/x_0) \times 100,$$

where x_0 and x_1 represent the values before and after the conditioning contraction, respectively, and $\Delta x\%$ is the adjusted logarithmic difference in percent.

Results

The repetitions analyzed for each parameter varied across sets and participants. This variability and some distinct patterns are shown in Figure 2. Changes in percent relative to the Pre values are shown in Figures 3(A) and 3(B).

Repetition of Maximum Peak TT

The 1-way analysis of variance revealed a significant main effect for the peak TT ($F_{4,42} = 168.6$, $P < .001$) with the post hoc test showing a significant increase in all sets (Set1–4) compared with Pre ($P < .001$, $g = 1.89$ – 2.23 , large; Table 1). Similarly, for RTD values the main effect was significant ($F_{4,42} = 103.0$, $P < .001$), while post hoc test showed a significant increase between sets (Set1–4) compared with Pre ($P < .001$, $g = 2.71$ – 3.12 , large). Furthermore, the peak TT for Set3 ($P < .05$, $g = 0.36$, small) and Set4 ($P < .001$, $g = 0.48$, small) was significantly lower compared with Set1. The main effect for TtP was significant ($F_{4,42} = 54.3$, $P < .001$) too. The post hoc analysis demonstrated a significant decrease for all sets compared with Pre ($P < .001$, $g = 1.68$ – 2.38 , large). A significant decrease in TtP was observed when comparing Set1 with Set2 ($P < .05$, $g = 0.76$, medium), Set3 ($P < .05$, $g = 0.80$, large), and Set4 ($P < .001$, $g = 0.91$, large). For the HRT, the analysis showed that the main effect was significant ($F_{4,42} = 26.5$, $P < .001$), with the post hoc analysis revealing a significant decrement of the values on the Set2, Set3, and Set4 relative to Pre ($g = 1.06$ – 1.52 , large) or Set1 ($P < .05$, $g = 0.56$ – 0.90 , medium to large; Table 1).

The SOL $M_{\max}$ ($F_{4,42} = 6.5$, $P < .001$), as well as the LG $M_{\max}$ ($F_{4,42} = 8.0$, $P < .001$), were significantly different between sets. The decrement in the SOL muscle, was significant during the Set3 ($P < .05$, $g = 0.33$, small) and Set4 ($P < .001$, $g = 0.43$, small) when compared with Pre and during Set4 compared with Set1 ($P < .05$, $g = 0.36$, small). On the other hand, LG demonstrated an increment

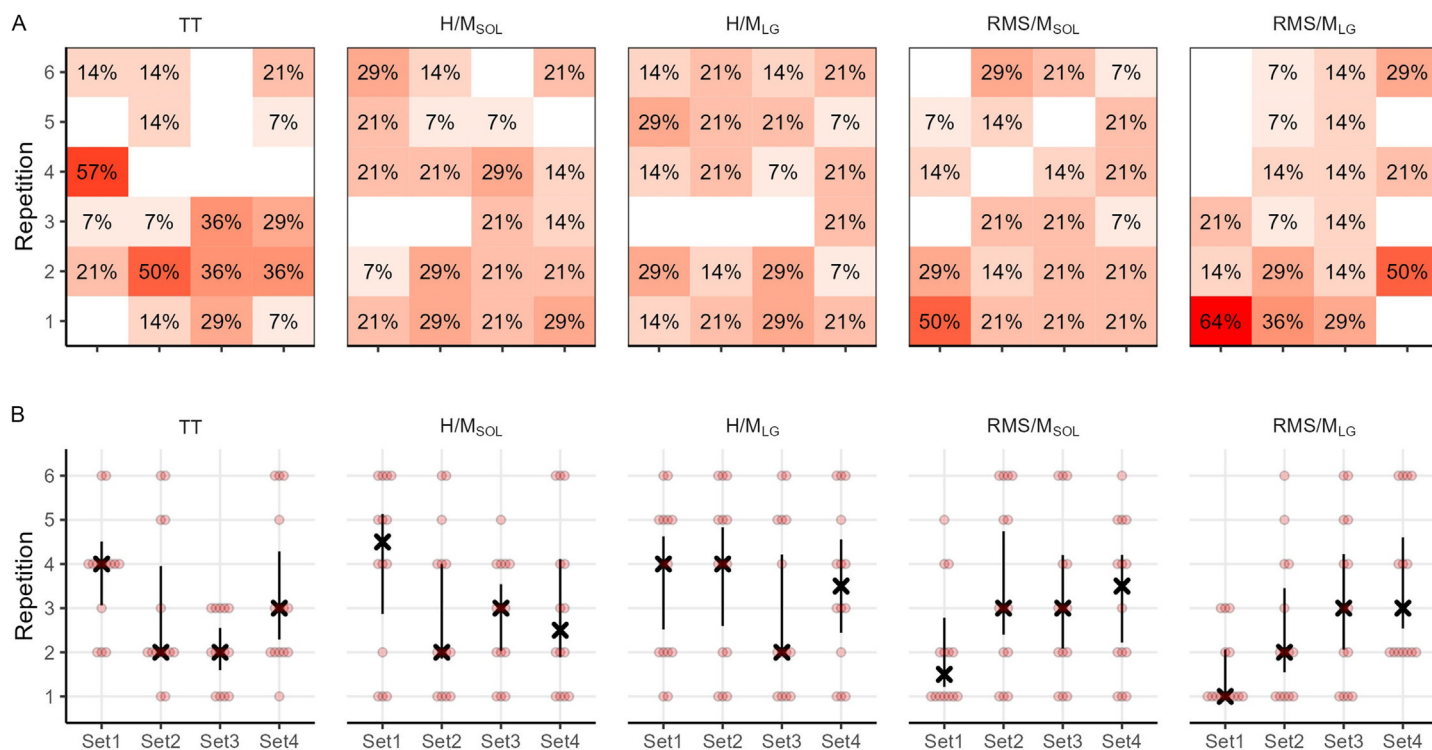


Figure 2 — (A) Heat map of the percentage of participants for each repetition and set, when the maximum value of the following parameters was recorded: peak TT, H/M, and RMS/M for the SOL and LG muscles. (B) Median value of the repetition number with the maximum value from each set. Vertical lines illustrate the 95% CI, and superimposed cycles represent individual participants. H/M indicates the ratio of H-reflex to maximal M-wave; LG, lateral gastrocnemius; RMS/M, root-mean-square electromyogram normalized to maximal M-wave; SOL, soleus; TT, twitch torque.

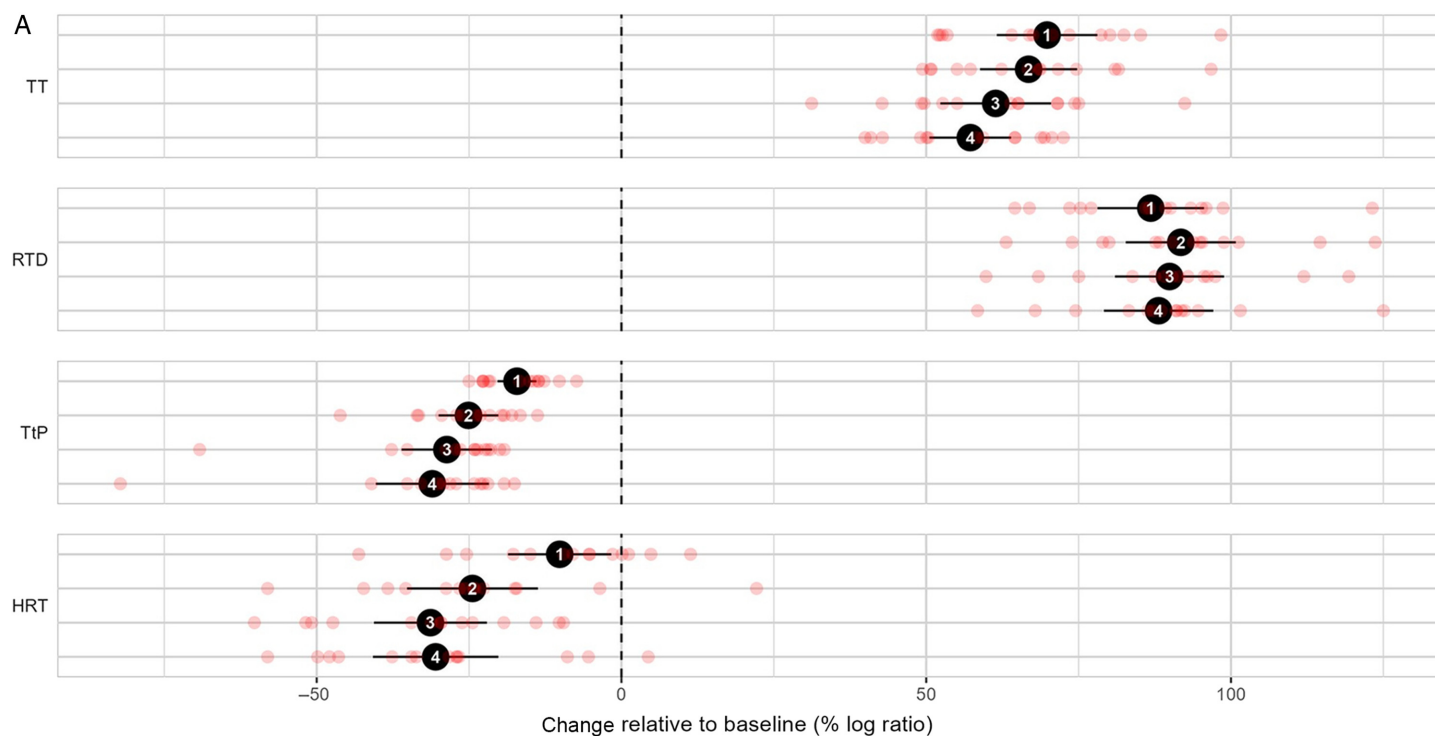


Figure 3(A) — Mean change (%log ratio) for each set of the muscular variables—peak TT, RTD, TtP, and HRT—relative to the baseline values before the conditioning contractions. Numbers in circles represent the set number, horizontal lines illustrate the SD around the mean, and smaller, shaded circles represent the values of each participant. TT indicates twitch torque; RTD, rate of torque development; TtP, time to peak torque; HRT, half relaxation time.

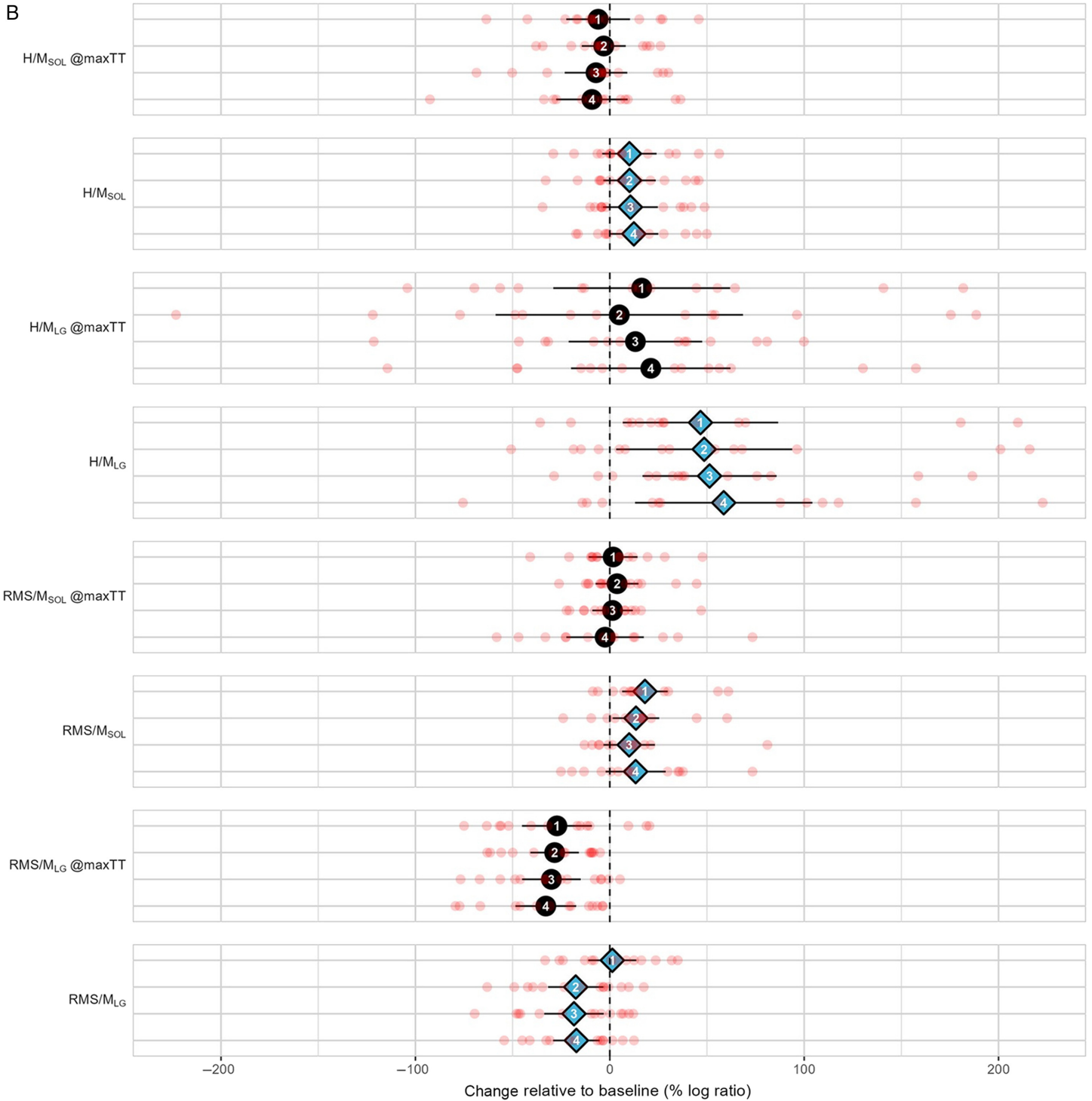


Figure 3(B) — Mean change (%log ratio) for each set of the neural variables—H/M and RMS/M for the SOL and LG muscles—relative to the baseline values before the conditioning contractions. Numbers in circles represent the set number, horizontal lines illustrate the SD around the mean, and smaller, shaded circles represent the values of each participant. Means in black and variables marked with “@maxTT” indicate that the repetition of the set that was evaluated was the one for which peak TT was at its maximum. Diamond-shaped squares indicate the mean of the repetition with the maximum value from each set. TT indicates twitch torque; H/M, ratio of H-reflex to maximal M-wave; SOL, soleus; LG, lateral gastrocnemius; RMS/M, root-mean-square electromyogram normalized to maximal M-wave.

during the Set1 compared with Pre ($P < .05$, $g = 0.25$, small) and a decrease during the Set3 ($P < .001$, $g = 0.26$, small) and Set4 ($P < .001$, $g = 0.31$, small) compared with Set1. H/M_{SOL} ($F_{4,42} = 0.40$, $P = .806$) as well as H/M_{LG} ($F_{4,42} = 0.89$, $P = .474$) did not change for all sets compared with Pre.

The 1-way analysis of variance revealed significant main effect for RMS/M_{LG} ($F_{4,42} = 8.13$, $P < .001$) but not for RMS/M_{SOL} ($F_{4,42} = 0.10$, $P = .981$). More specifically for the LG, the values of all sets (Set1–4) compared Pre were significantly lower ($P < .001$, $g = 0.06$ – 0.33 , negligible to small; Table 1).

The values of VA were significantly different between sets ($F_{3,39} = 4.29$, $P = .010$), and the post hoc test showed a significant decrease in Set4 compared with Set1 ($P < .05$, $g = 0.70$, medium).

Repetition of Maximum Value Within Each Set

H/M_{SOL} did not change through time ($F_{4,42} = 1.9$, $P = .111$). In contrast, H/M_{LG} demonstrated a significant main effect ($F_{4,42} = 4.3$, $P < .05$) and revealed an increment for all the sets (Set1–4) compared with Pre ($g = 0.28$ – 0.42 , small; Table 2). Furthermore, the 1-way analysis of variance revealed a significant main effect for RMS/M_{SOL} ($F_{4,42} = 2.6$, $P < .05$) and RMS/M_{LG} ($F_{4,42} = 5.2$, $P = .001$). More specifically, the post hoc test showed that RMS/M_{SOL} increased significantly during Set1 compared with Pre ($P < .05$, $g = 0.05$, negligible). The RMS/M_{LG} revealed a significant decrease in Set2–4 compared with Pre ($g = 0.53$ – 0.54 , medium; Table 2). No difference between the Sets was observed for VA ($F_{3,39} = 0.4$, $P = .773$).

The TT was significantly different between the sets when selecting the repetitions of maximum value in H/M_{SOL} ($F_{4,42} = 101.2$, $P < .001$, $g = 1.71$ – 2.06 , large), H/M_{LG} ($F_{4,42} = 95.4$, $P < .001$, $g = 1.72$ – 2.05 , large), RMS/M_{SOL} ($F_{4,42} = 105.9$, $P < .001$, $g = 1.69$ – 2.08 , large), RMS/M_{LG} ($F_{4,42} = 100.8$, $P < .001$, $g = 1.74$ – 2.07 , large), and VA ($F_{4,42} = 100.5$, $P < .001$, $g = 1.73$ – 2.03 , large). Post hoc test showed a significant increase for all the above variables on all sets (Set1–4) compared with Pre (Table 2).

Discussion

This study verified the presence of TT potentiation after a conditioning stimulus and the gradual decrease of this potentiation due to the repetitive isometric contractions and the progression of fatigue, which lies on the opposite spectrum of PAP. Similar but not identical behavior was observed for RTD, TtP, and HRT. Regarding the H-reflex, the background muscle activation prevented any

Table 1 Mean (SD) of All Participants for All Variables Regarding the Examined TT and Neural Components, Before the Conditioning (Pre) and During Each Set (Set1, Set2, Set3, and Set4)

	Pre	Set1	Set2	Set3	Set4
Peak TT, N·m	16.3 (5.2)	32.2 (8.3)**	31.2 (7.9)**	29.4 (6.6)**†	28.4 (7.0)**††
RTD, N·m/s	172.8 (55.1)	405.7 (103.8)**	426.0 (108.43)**	415.8 (91.7)**	412.5 (108.3)**
TtP, ms	94.6 (9.3)	79.7 (7.8)**	73.6 (7.6)**†	71.7 (11.1)**†	70.3 (11.9)**††
HRT, ms	83.1 (13.6)	75.9 (16.6)	66.1 (17.2)**†	61.5 (14.4)**††	61.8 (13.5)**††
SOL M _{max} , mV	15.0 (3.1)	14.7 (3.3)	14.0 (3.9)	13.8 (3.8)*	13.3 (4.1)**†
LG M _{max} , mV	11.4 (3.8)	12.4 (3.6)*	11.8 (3.7)	11.4 (3.6)††	11.3 (3.5)††
H/M _{SOL}	0.584 (0.226)	0.546 (0.197)	0.561 (0.208)	0.563 (0.238)	0.552 (0.213)
H/M _{LG}	0.062 (0.036)	0.081 (0.051)	0.085 (0.076)	0.076 (0.060)	0.079 (0.055)
RMS/M _{SOL} , %	2.96 (0.70)	3.00 (0.61)	3.07 (0.68)	2.99 (0.68)	2.96 (1.01)
RMS/M _{LG} , %	5.96 (2.73)	4.37 (1.36)**	4.56 (2.28)**	4.55 (2.43)**	4.52 (2.56)**
VA, %	—	95.9 (1.4)	96.0 (1.7)	97.3 (0.7)	90.5 (2.5)*

Abbreviations: H/M, ratio of H-reflex to M_{max}; HRT, half relaxation time; LG, lateral gastrocnemius; M_{max}, maximal M-wave; RMS/M, root-mean-square electromyogram normalized to M_{max}; RTD, rate of torque development; SOL, soleus; TT, twitch torque; TtP, time to peak torque; VA, voluntary activation. Note: The trial of each set that was selected for analysis was the one in which twitch torque was maximal.

Significant difference (* $P < .05$, ** $P < .001$) compared with Pre. Significant difference († $P < .05$, †† $P < .001$) compared with Set1.

Table 2 Mean (SD) of All Participants for All Variables Regarding the Examined Neural Components, Before the Conditioning (Pre) and During Each Set (Set1, Set2, Set3, and Set4)

	Pre	Set1	Set2	Set3	Set4
H/M _{SOL}	0.584 (0.226)	0.625 (0.185)	0.625 (0.181)	0.644 (0.206)	0.649 (0.197)
H/M _{LG}	0.062 (0.036)	0.102 (0.017)*	0.107 (0.021)*	0.105 (0.017)*	0.112 (0.073)*
RMS/M _{SOL} , %	2.96 (0.70)	3.52 (0.69)*	3.38 (0.75)	3.31 (1.04)	3.42 (1.05)
RMS/M _{LG} , %	5.96 (2.73)	5.81 (1.92)	5.08 (2.58)*	5.05 (2.61)*	5.15 (0.2.81)*
VA, %	—	99.8 (1.5)	99.8 (0.8)	99.7 (1.1)	99.4 (1.3)
[H/M _{SOL}] ^a , N·m	16.3 (5.2)	29.7 (8.0)**	29.3 (7.2)**	28.7 (6.4)**	26.6 (6.5)**
[H/M _{LG}] ^a , N·m	16.3 (5.2)	29.7 (7.4)**	29.1 (7.3)**	27.7 (6.4)**	26.6 (6.4)**
[RMS/M _{SOL}] ^a , N·m	16.3 (5.2)	29.4 (7.0)**	30.0 (8.2)**	28.2 (7.0)**	27.1 (7.1)**
[RMS/M _{LG}] ^a , N·m	16.3 (5.2)	28.9 (7.7)**	29.4 (7.0)**	28.0 (6.4)**	27.2 (6.9)**
[VA] ^a , N·m	16.3 (5.2)	30.6 (8.5)**	30.6 (8.2)**	28.2 (7.4)**	27.4 (7.1)**

Abbreviations: H/M, ratio of H-reflex to maximal M-wave; LG, lateral gastrocnemius; RMS/M, root-mean-square electromyogram normalized to maximal M-wave; SOL, soleus; VA, voluntary activation. Note: For each variable, the maximum value from each set was included in the analysis.

^aTwitch torque (in newton meters) during the repetition of each set when the maximum value was observed for the variable shown in square brackets.

Significant difference (* $P < .05$, ** $P < .001$) compared with Pre.

inhibition after the conditioning MVICs, but this was not enough to reveal any systematic facilitation. However, LG H-reflex was facilitated compared with the baseline values, but not for the repetitions within the sets when the TT was maximal. Similar findings have been shown for the other neural parameters too, indicating that muscular and neural mechanisms act in an asynchronous manner during an isometric protocol, when PAP and fatigue coexist.

The implementation of the current protocol with 6 seconds of successive MVICs and the evaluation of neuromuscular properties during the 15-second interval between the repetitions, give insights into the process of potentiation and the counteraction of fatigue. In the present study the maximal TT potentiation was observed during Set1, and in most participants (57%) this was observed during the fourth repetition (Figure 1). This implies that if the goal is to maximize TT, a single set with 4 contractions could be probably ideal. This finding is close to previous studies, testing 5-second isometric knee extensions with 3-second rest in between (~fourth repetition)¹⁹ or dorsiflexions with 3-second duration and 5-second rest (~third repetition).²⁶ The existence of repeated isometric contractions and their cumulative effect on TT potentiation, may also explain the higher potentiation observed in the present study compared with others that used a single MVIC as conditioning stimulus.^{8,10,19,27}

Alongside with the TT potentiation, we observed changes in RTD, TtP, and HRT. Similar studies^{18,28} have also reported parallel changes in TT and RTD, and this could be explained by the fundamental mechanism of PAP, which is attributed to the phosphorylation of myosin regulatory light chains that affects the light chain kinase via enhancing the Ca^{2+} sensitivity to the actin-myosin complex.²⁹ This results in increasing the rate of cross-bridge attachments, with less required Ca^{2+} , which potentially increases RTD. The shortened TtP and HRT have been previously reported after a maximum conditioning contraction^{3,13} and are more likely linked to the kinetics of the sarcoplasmic reticulum.³⁰ Based on the evidence from the current study, we could claim that TT potentiation is attributed at least partly to mechanisms that act at peripheral level. However, it has to be noted that the fatigue effect observed in TT during the Set3 and Set4 when compared with Set1 (Table 1 and Figure 2A), was not apparent in RTD, which remained unchanged, or in TtP and HRT, which were decreased (ie, improved) even further. This pinpoints that these properties are affected differently by fatigue, an observation that is in agreement with a previous study at least for the TtP and HRT of the knee extensors.¹⁹

Regarding the spinal reflex excitability assessed with H-reflex and the muscle activation assessed with RMS/M and VA, there was no indication for facilitation/increase at the repetition within the set corresponding to maximal TT potentiation (Table 1). On the contrary, the LG muscle, which is more prone to fatigue than the SOL,³¹ demonstrated a decreased RMS/M from the first to the last set, and VA was lower at Set4 compared with Set1, indicating muscle-specific signs of central fatigue.³² However, analyzing the same variables for the repetition when reaching the maximum value within a set, an increase in H/M_{LG} and $\text{RMS/M}_{\text{SOL}}$ was observed. From Figure 1, it is clear that the repetition within the set corresponding to the maximal observed value, is different between the parameters, and highly variable for the H/M. This suggests that the temporal component of analyzing neural variables should be considered for each variable separately. Furthermore, future studies should tackle the source of variability between subjects especially for the neural variables. This might give a clearer picture especially on the temporal domain.

Interestingly, in previous studies reporting an increment in H-reflex after the conditioning contraction,^{6–8} this increment was derived from the highest H-reflex value that occurred during the recovery period (3–11 min), which does not coincide with the time corresponding to maximal TT potentiation. The current study attempted to assess the H-reflex within 15 seconds after the end of each MVIC (when PAP is present) and reported no significant change, in contrast to previous studies that reported an inhibition in H-reflex,^{6,7,10} which is attributed possibly to postactivation depression.¹⁰ A key difference in our approach was that the H-reflex was assessed during a low level of VA, which reduces the presence of PAD^{10,33} and may reveal any H-reflex potentiation attributed to the conditioning contraction.

The current study is the first one that reported facilitation instead of inhibition on the H/M_{LG} , 12 seconds after the MVICs (even with a small effect size), for the repetition corresponding to the maximal observed value within each set. For the same repetitions, TT was also facilitated, in all sets, suggesting that neural facilitation and TT potentiation may coexist (Table 2). It should also be noted that the stimulation intensity of the H-reflex recordings was optimized for the SOL, but not the LG muscle. Therefore, the increase in H/M_{LG} should be considered with caution and further studies targeting explicitly the investigation of LG should be designed in the future to make this outcome more solid. The increase in H/M_{LG} and lack of any change in H/M_{SOL} could be explained considering that LG has higher proportion of fast-twitch muscle fibers compared with SOL,³⁴ and is more prone to PAP.³ It could also be argued that when we analyze the highest value of H-reflex within each set, the result might be artificially inflated in the direction of H-reflex facilitation, given that a specific repetition within each set was explicitly selected for its maximal value. This could possibly partially explain the observed facilitation of H-reflex in earlier studies that evaluated the highest H-reflex amplitude within 5 minutes after the conditioning stimulus,⁸ in contrast with others that used a different approach and did not find any increase.³³

Regarding the potential link of our findings with postactivation performance enhancement (PAPE), a term known for increasing voluntary performance that takes several (~5) minutes to appear,^{35,36} we are not confident that neural factors may contribute to this phenomenon for the following reasons. Although the TT properties were at potentiated state for at least one repetition during each set, the statistically significant increase in H/M_{LG} and $\text{RMS/M}_{\text{SOL}}$ that may coincide with the expected appearance of PAPE, had a small and negligible effect size, respectively. Furthermore, a decrease in RMS/M_{LG} and VA was observed, which advocates for absence of PAPE while fatigue emerges. On the other hand, the extent that the potentiated properties of TT may play an important role in PAPE, remains to be seen with laborious measurement setups, reassuring that performance is measured only once, after each variant of different conditioning stimulation protocols.

Practical Applications

The present study shows that a single set of 4 contractions is, in most cases, ideal for revealing the maximum TT potentiation. This implies that it is more likely that further trials could have a decreasing effect on TT. However, other neural factors that could affect performance have a different time (repetition of set) of peak response, when TT potentiation is not necessarily at its peak. This indicates that neural factors should not be accounted for TT potentiation, although it cannot be excluded that they might contribute to PAPE that is reported after PAP.^{35,36} Nevertheless,

fatigue and PAP counteract to each other,³⁷ that is, the presence of fatigue may mask the occurrence of PAP or vice versa. Therefore, our findings are specific to the current protocol, and hence any generalization of these outcomes should be made with caution. Moreover, for sports scientists that evaluate the H-reflex after a conditioning stimulus, this study suggests H-reflex recordings with a low-level background voluntary contraction of the muscle of interest may reveal a potential enhancement, by reducing the inherent postactivation depression. This could contribute to fewer disparities among studies related to this issue. Finally, researchers should consider the variability of potentiation between participants, especially in the neural-related variables, which implies the possible need for individualized approaches.

Conclusion

For the examined protocol consisting of 4 sets of 6 MVICs with 15-second intervals between the repetitions and 2-minute rests between the sets, we observed no neural facilitation between the repetitions when the twitch torque was at its maximum. This indicates that the phenomenon of postactivation potentiation (PAP) may not be attributed to the examined neural factors. However, the maximum within each set H/M_{LG} , with background EMG activity, is facilitated in contrast to the inhibition that was previously reported without background EMG activity. This indicates that neural factors might contribute to postactivation performance enhancement but not PAP. To clarify this, future experiments regarding the examination of muscular and neural mechanisms of PAP and fatigue should consider their asynchronous existence concerning the appearance of their maximal values and their inherent between-participants variability.

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